

Drishti Kavach

**Advancing Indian Railways Security System with Spatial Clearance
Perception Engine**





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01

Introduction



Abstract

Drishti-Kavach is an edge-deployable optical perception system designed to provide forward-looking track clearance verification for Indian Railways' Kavach (TCAS) network. It uses a decoupled dual-model vision pipeline combining **BiSeNetV2** for real-time track and rail segmentation with **YOLO11m** for 8-class obstacle and sabotage detection. The system applies vector spatial geometry to classify threats into three distinct safety levels (**Critical, Warning, Safe**) and incorporates active **850nm Near-Infrared night vision** along with adaptive defogging algorithms to maintain reliable, real-time operation at over 50 FPS in all weather conditions.



Problem Statement

India's railways face recurring accidents due to human error, signal violations, and unexpected track intrusions. While Kavach addresses train-to-train collision prevention, it lacks the capability to visually detect and classify foreign obstacles such as people, animals, or vehicles on the track in real time. Existing track monitoring solutions are either too costly, infrastructure-heavy, or not accessible for rapid prototyping and demonstration. Therefore, there is a need for an affordable, intelligent system that combines AI based visual detection with IoT-driven train control to demonstrate and extend the capabilities of existing railway safety infrastructure like Kavach.



Why We Chose this Project

- **Indian Railways** is one of the **world's largest railway networks**, spanning over **69,000 route kilometres** across India, connecting major cities, towns and remote regions.
- It carries approximately **18.87 million passengers every day**, making railways a critical mode of mass transportation in India.
- In 2023-24, Indian Railways recorded around 690.5 crore passenger journeys and earned approximately **₹70,693 crore from passenger services**.
- Indian Railways generated ₹2.56 lakh crore in gross earnings during 2023-24, with ₹3,260 crore in net revenue, demonstrating its enormous economic importance.



Why we chose this project

- Despite this scale, railway safety remains a critical concern. **313 people were killed in 40 consequential train accidents during 2023-24.**
- **2025 - Jalgaon:** Around 12 people were killed after passengers who had exited the Pushpak Express onto the adjacent track following a false fire alarm were struck by the Karnataka Express, highlighting the danger of human presence/obstacles on tracks.
- **2024 - Kanchanjunga Express:** A goods train collided with the Kanchanjunga Express near Rangapani, West Bengal, killing 11 people and injuring 60+, demonstrating the consequences of rear-end collisions and signalling/operational failures.
- **2023 - Balasore:** The triple-train accident involving the Coromandel Express, Bengaluru-Howrah Express and a freight train killed 296 people, with a signalling error identified as the key factor.



Why we chose this project

- **2016 - Indore-Patna Express:** The train derailed near Pukhrayan, Uttar Pradesh, killing approximately 150 people; a fractured rail was identified as a major contributing factor.
- **2010 - Jnaneswari Express:** The train derailed near Jhargram, West Bengal, resulting in 148+ deaths, with suspected sabotage involved.
- **2002 - Rafiganj:** The Howrah-New Delhi Rajdhani Express derailed over the Dhawe River in Bihar, killing 130+ people; sabotage was suspected.
- Although railway safety has improved significantly - with consequential accidents falling from 473 in 2000-01 to 40 in 2023-24 and accidents per million train-km falling by over 95% - the persistence of collisions, derailments, signalling failures and track-related hazards highlights the need for real-time detection and automated preventive safety systems such as Drishti Kavach.



Introduction

- Railway safety is a critical concern in India, where the network serves over 13 million passengers daily.
- Modern technologies like Artificial Intelligence and IoT offer new possibilities to enhance traditional safety systems.
- Kavach is India's indigenous Automatic Train Protection system designed to prevent train collisions and signal violations.
- The main aim is to demonstrate how AI-based obstacle detection and IoT-driven train control can work together to make railways safer.



→ **SIL - 4 Certified**

Built to the Highest International Safety Certification!

→ **23,360 Kilometers**

Current track-side implementation record!

→ **₹ 2,763.90 Crores**

Total utilized expenditure on Kavach Implementation!





Objectives

- **Automated Spatial Awareness:** To implement a vision-based system that autonomously identifies the physical boundaries of railway tracks (the “Danger Zone”) from a stationary trackside perspective.
- **Real-Time Intrusion Detection:** To detect and classify physical obstacles (humans, animals, or inanimate) that enter the track area in real-time.
- **Establish a Scalable Foundation:** For future enhancements including low-light detection, fog and rain resilience, and integrating with physical IoT alerting hardware.
- **Safety Integrity Level (SIL) Enhancement:** To provide a supplementary layer of visual data to the Kavach 4.0 system, moving beyond purely signal-based protection to physical environment monitoring.



02

Literature Survey





Literature Survey

YEAR	PAPER TITLE	METHODOLOGY	ADVANTAGES & DISADVANTAGES	LIMITATIONS	FUTURE SCOPE
2025	Real-Time Railway Obstacle Detection Based on Multitask Perception Learning, C. Chen et al. (BASE PAPER)	End-to-end MTP-Rail model performing simultaneous object detection and track segmentation.	• High classification accuracy (89.6%) by analyzing spatial relationships. • Significant computational demands for the "Large" model variant.	High resource consumption makes it difficult for edge hardware.	Optimization for deployment on ultra-low-power onboard devices.
2026	YOLO-Rail: An Improved YOLO Model for Obstacle Detection, Z. Wang & X. Du	Integrates ConvNeXtV2 modules and depth wise separable convolutions into a lightweight YOLO model; pre-trained on the COCO dataset.	• Optimized for real-time onboard processing; superior inference speed. • Requires extensive fine-tuning to avoid over-fitting to specific regions.	Risk of over-fitting to specific track types or environments.	Inclusion of rare hazards like landslides or flooding in datasets.
2025	Track2Net: A Fast Lightweight Model with Keypoint Alignment and Track Anchors, H. Guo et al.	Uses Keypoint Alignment for simple scenes and CTPM for tunnels/turnouts.	• High efficiency (243 FPS) and robust accuracy (91.43%) in high-speed scenarios. • Performance is	Accuracy can be negatively affected by train vibrations.	Exploring adaptability under different speeds and track types.



Literature Survey

YEAR	PAPER TITLE	METHODOLOGY	ADVANTAGES & DISADVANTAGES	LIMITATIONS	FUTURE SCOPE
2025	Improved RT-DETR Framework for Railway Obstacle Detection ,P. Li et al.	Introduces Multiscale Separable Deformable (MSD) and Deformable Agent Attention (DAA).	- Reduces parameters by 24.6%; achieves high detection speeds of 90 FPS. - Single-stage implementation struggles with extremely small objects.	Accuracy issues with tiny objects due to model redundancy reduction.	Deploying MSD and DAA modules into broader domains like UAVs.
2025	RAE3D: Multiscale Aggregation-Enhanced 3D Object Detection, L. Lian et al.	Single-stage architecture with MVA-PQ and BEV-MLA modules for voxel aggregation.	- Effectively handles sparse long-range LiDAR point clouds. - LiDAR-based methods are sensitive to heavy rain or thick fog.	Point cloud quality degrades significantly in adverse weather.	Optimizing hybrid voxel-point representation for mobile platforms.
2025	RailBEV: A Vision-Based Bird's-Eye-View Representation Network, W. Liu et al.	Uses Depth Supervision View Transformation (DSVT) to project images to BEV plane.	- Enhances long-range detection (>120m) using dense depth labels. - Vision-based depth accuracy is lower than	Lower precision for near-field objects compared to direct sensors.	Integrating temporal frames for tracking and velocity prediction.



Literature Survey

YEAR	PAPER TITLE	METHODOLOGY	ADVANTAGES & DISADVANTAGES	LIMITATIONS	FUTURE SCOPE
2025	Leveraging Domain-Specific Features for High-Performance Obstacle Classification, L. Eisentraut et al.	Integrates a Gaussian filter into ResNet-50 to suppress high-frequency noise (gravel).	• High balanced accuracy (99.26%); proves domain filtering beats complexity. • Focused primarily on classification rather than precise localization.	Lacks bounding box detection capabilities for spatial awareness.	Implementing the filter into broader object detection frameworks.
2025	Multi-AGV Rail Transportation Path Planning and Dynamic Obstacle Avoidance, Z. Wu & T. Li	Combines A* algorithm with deep reinforcement learning (GG-DRL).	• Reduces collisions by 70%; decreases planning time by 25-30%. • Heuristic search complexity increases in dense, large-scale environments.	Scalability issues when managing a massive number of agents.	Applying GG-DRL to automated warehouse and logistics systems.
2025	Advancements in OID Methods for Rail Transit: A Comprehensive Review, X. Xu et al.	Summarizes LiDAR, Camera, and Radar schemes and AI-based algorithms.	• Provides a structured roadmap; highlights shift to deep learning. • Identifies a domain gap	Synthetic training data does not always match real-world criticality.	Exploring foundation models (e.g., CLIP) for zero-shot hazard detection.



Literature Survey

YEAR	PAPER TITLE	METHODOLOGY	ADVANTAGES & DISADVANTAGES	LIMITATIONS	FUTURE SCOPE
2025	BDANet: Boundary-Assisted Dual-Branch Attention Semantic Segmentation, X. Song et al.	Boundary-assisted decoder with Global-Local Attention (GLAM) for wider receptive fields.	• Enables real-time recognition (35.83 FPS); uses unsupervised domain adaptation. • COCO-Stuff-Rail dataset lacks extreme weather samples.	Difficulty adapting to heavy rain or snow without manual labeling.	Scaling for diverse environmental distributions and open-set detection.



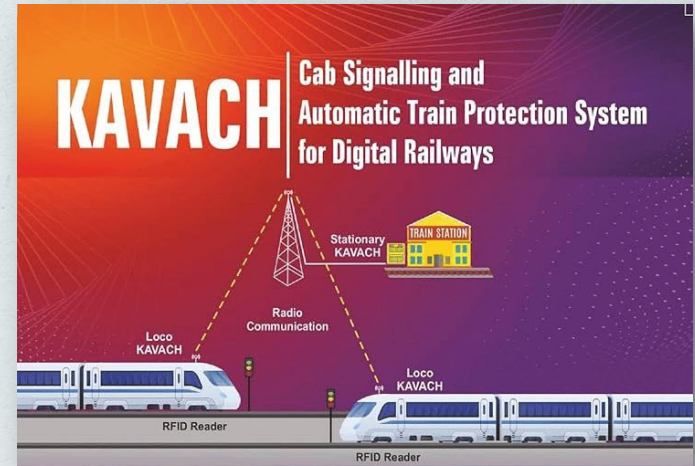
03

Existing & Proposed System



Existing System

- Kavach is India's Automatic Train Protection system that prevents train collisions and signal violations.
- It uses RFID balises and radio communication between the locomotive and trackside units.
- Automatically applies brakes when over speeding or approaching a red signal.
- Does not have any camera-based detection to identify obstacles on the track.
- Expensive to deploy and currently limited in coverage across the Indian railway network.



Existing System

Advantages:

1. Effectively prevents train-to-train collisions and signal violations.
2. Fully automated with no dependency on human intervention.
3. Proven and tested technology adopted by Indian Railways officially.
4. Operates reliably in all weather and low visibility conditions.
5. Enables real-time communication between locomotive and trackside units.

Disadvantages:

1. Cannot detect visual obstacles like people, animals, or vehicles on track.
2. Very expensive to implement and maintain across the network.
3. Coverage is still limited to select routes across India.
4. Does not classify or identify the type of threat on the track.
5. Fully dependent on radio communication, vulnerable to signal interference.



Proposed System

- **Decoupled Dual-Model Architecture:** Combines BiSeNetV2 for real-time drivable ballast and rail segmentation with a customized YOLO11m detector for 8 railway hazard classes.
- **Deterministic Vector Spatial Reasoning:** Uses polygon intersection geometry to evaluate ground-contact footprints against track boundaries, categorizing threats into Critical, Warning, and Safe zones.
- **24/7 Multi-Spectral Vision:** Employs active 850nm Near-Infrared processing paired with adaptive defogging (CLAHE and Dark Channel Prior) for round-the-clock reliability in heavy fog, rain, and darkness.
- **Automatic Train Protection Integration:** Provides direct telemetry signals to locomotive braking interfaces while rendering a real-time driver assistance HUD.
- **Edge-Deployable Performance:** Achieves over 50 FPS with an end-to-end processing latency under 20 ms on locomotive-grade hardware.



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Methodology



Methodology

- **Dataset Harmonization and Multi-Spectral Synthesis:** Standardized RailSem19 and UAV-RSOD datasets into unified 3-class segmentation and 8-class detection schemas, paired with synthetic 850nm Near-Infrared transformations.
- **Atmospheric Optical Preprocessing:** Ingested video streams through dynamic fog density scoring, LAB-color CLAHE contrast enhancement, and Dark Channel Prior dehazing to restore visibility in severe weather.
- **Decoupled Dual-Engine Inference:** Executed BiSeNetV2 at 512x1024 for ballast and rail line segmentation alongside YOLO11m at 1024x1024 for high-resolution obstacle and sabotage detection.



Methodology

- **Deterministic Vector Spatial Reasoning:** Converted segmentation masks into Shapely polygons, projected bottom 25% ground contact footprints for obstacles, and evaluated spatial intersections with dynamic clearance buffers.
- **Safety Classification and Telemetry Output:** Assessed threat levels across Critical, Warning, and Safe categories to trigger automatic train protection braking signals and render real-time driver HUD overlays.

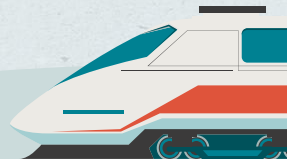


System Specifications



Hardware Components:

1. Forward-Facing Optical and 850nm NIR Night Vision Camera
2. Active 850nm Infrared Spotlight Illuminator
3. Embedded Edge AI Processing Unit (NVIDIA Jetson AGX Orin / Industrial GPU)
4. In-Cab Driver Assistance Display (HUD)
5. Locomotive Kavach (TCAS) ATP Braking Relay Interface

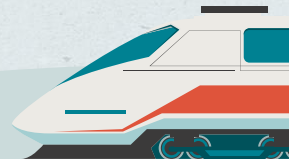


System Specifications



Software Components:

1. Python Runtime
2. PyTorch Deep Learning Framework
3. Ultralytics YOLO Library
4. BiSeNetv2 – Railway track segmentation
5. YOLO11 – Obstacle detection and classification
6. ONNX Runtime Engine
7. OpenCV Computer Vision Library
8. Shapely Vector Geometry Engine
9. NumPy Scientific Computing Package



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System Design



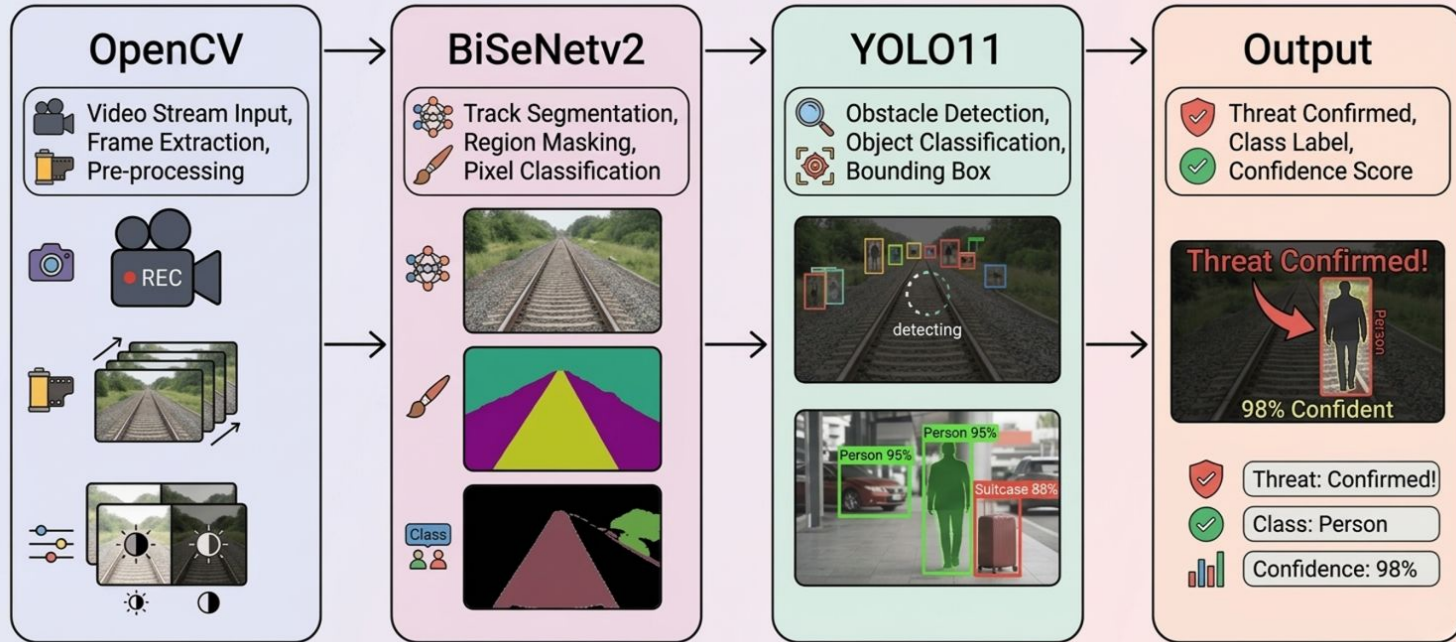


System Design

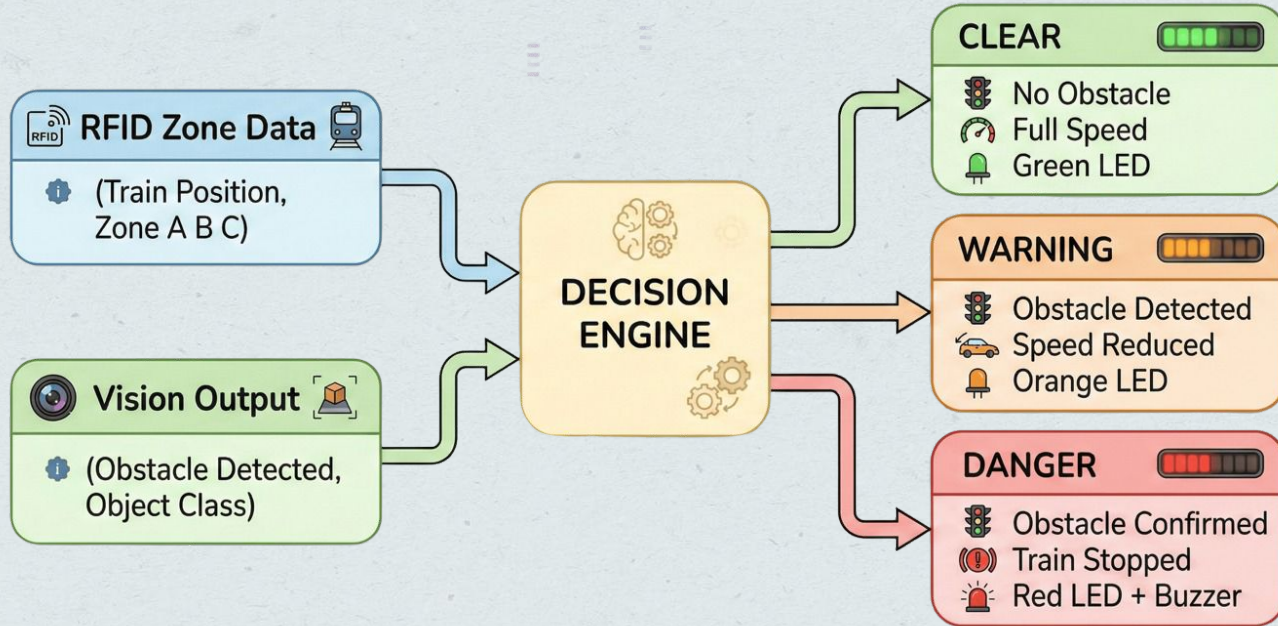
1. **Input Module:** Optical and 850nm Near-Infrared forward camera captures live 1080p track video feeds at 30 to 60 FPS, continuously delivering frames to the ingestion pipeline with minimal latency.
2. **Communication Module:** Video frames are passed to the inference pipeline through thread-safe buffers, with real-time telemetry state records written periodically to synchronized data files and IPC sockets for dashboard and locomotive sub-rack integration.
3. **Processing Module:** Performs atmospheric optical enhancement (CLAHE and Dark Channel Prior defogging) and runs parallel inference using BiSeNetV2 for track bed and rail segmentation alongside YOLO11m for 8-class hazard and sabotage detection.
4. **Decision Module:** Vector spatial engine projects the bottom 25% ground-contact footprint of each detected object and evaluates geometric polygon intersections against track boundaries, categorizing events into Critical, Warning, and Safe states.
5. **Output Module:** Renders a high-contrast driver assistance HUD with dynamic alert banners and distance overlays via OpenCV, while transmitting real-time telemetry signals to the locomotive Kavach Automatic Train Protection interface.



Processing Module



Decision Module

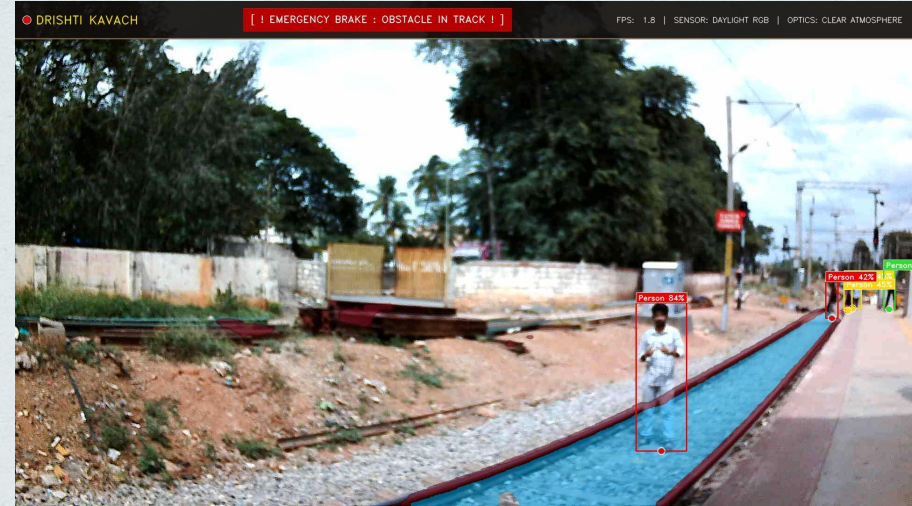
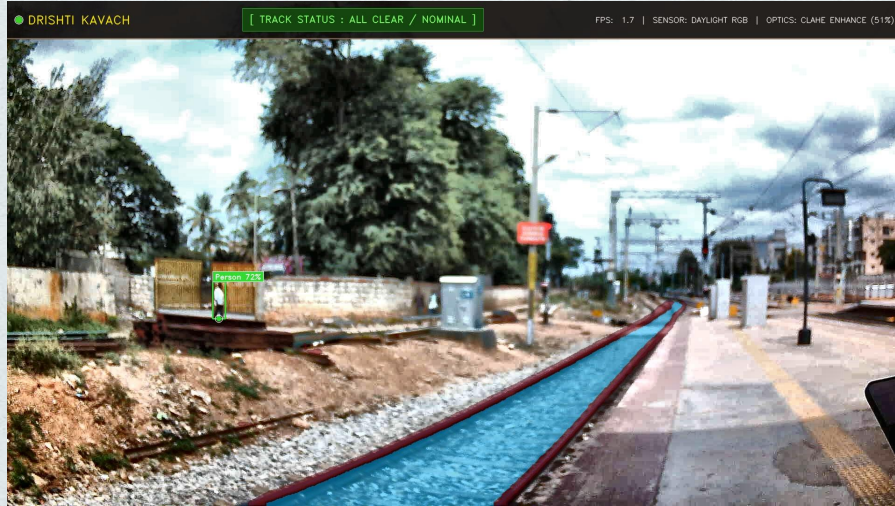


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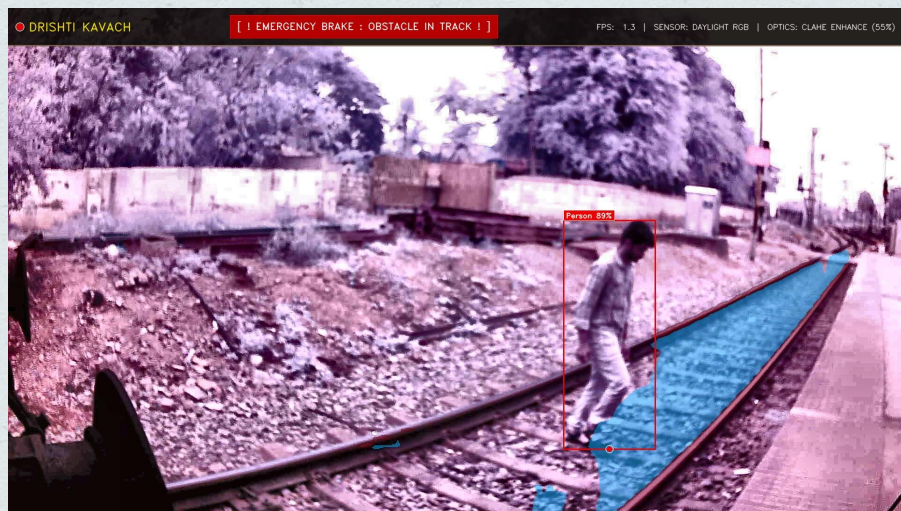
Results



Drishti Kavach v1.0 – Test Run Outputs



Drishti Kavach v1.0 – Test Run Outputs



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References



References

- [1] C. Chen, et al., *"Real-Time Railway Obstacle Detection Based on Multitask Perception Learning"*, 2025.
- [2] Z. Wang and X. Du, *"YOLO-Rail: An Improved YOLO Model for Obstacle Detection"*, 2026.
- [3] H. Guo, et al., *"Track2Net: A Fast Lightweight Model with Keypoint Alignment and Track Anchors"*, 2025.
- [4] P. Li, et al., *"Improved RT-DETR Framework for Railway Obstacle Detection"*, 2025.
- [5] L. Lian, et al., *"RAE3D: Multiscale Aggregation-Enhanced 3D Object Detection"*, 2025.
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- [10] X. Song, et al., *"BDANet: Boundary-Assisted Dual-Branch Attention Semantic Segmentation"*, 2025.



Thank You!

